OIPPP – Unit Test Report

Issue: #86 – Unit Test Report
Project: Online Internship & Placement Preparation Portal (OIPPP)
Test date: 4 November 2026
Branch: test/issue-86-unit-test-report
Tested commit: e74b47bd9d8031ff5d3b93371b64b873b3849488
Environment: Windows 11; Node.js 22.22.2 (client); npm 10.9.7 (client); Jest; Vitest 5.0.5.

1. Executive summary

The server and client automated test suites completed successfully.

Component	Suites / files	Passed	Failed
Server (Jest)	40	486	0
Client (Vitest)	6	53	0
Total	**46**	**539**	**0**

Overall result: PASS – 539 of 539 tests passed.

2. Server test results

Command: npm test -- --runInBand (from server/)
40 suites passed; 486 tests passed; 0 failed; 0 snapshots; duration 173.709 seconds.

Server coverage – npm run test:coverage

Metric	Coverage
Statements	95.16% (1,241 / 1,304)
Branches	89.06% (554 / 622)
Functions	94.90% (242 / 255)
Lines	95.99% (1,127 / 1,174)

Coverage run: 40 suites and 486 tests passed (0 failed); duration 192.917 seconds.

3. Client test results

Command: npm test -- --run (from client/)
6 test files passed; 53 tests passed; 0 failed; duration 33.56 seconds; Vitest 5.0.5. The tests cover API client behavior, route/role guards, page flows, validation, timer behavior, and status formatting.

Environment note: The initial attempt under Node.js 20.19.5 could not start Vitest workers because the installed dependencies require a newer runtime. Node.js 22.22.2 was used for the successful rerun.

4. Requirement-to-test mapping

Requirement area	Automated evidence	Result
Authentication / identity	auth.test.js; identity-flow.test.js; auth-service.test.js; auth-validation.test.js	Pass
Profile / resume	profile.test.js; resume.test.js; profile-model.test.js; user-profile-schema.test.js	Pass
RBAC / security	rbac.test.js; audit.test.js; identity-edge-cases.test.js; validate-middleware.test.js; auth-middleware.test.js; identity-infrastructure.test.js	Pass
Listings / search	listings.test.js; listing-search.test.js; listing-recommendations.test.js; bookmarks.test.js; listing-model.test.js; listing-service.test.js	Pass
Applications / review	apply.test.js; review.test.js; status.test.js; notifications.test.js; application-transitions.test.js	Pass
Aptitude / recovery	aptitude-api.test.js; aptitude-recovery.test.js; aptitude-indexes.test.js; aptitude-models.test.js; aptitude-scoring.test.js; attempt-timeout-job.test.js;	Pass
Analytics	analytics.test.js; analytics-performance.test.js	Pass
Reliability / supporting APIs	reliability.test.js; health.test.js; request-log.test.js; settings.test.js; admin.test.js	Pass

Scope note: SEC-2 deployed HTTPS/TLS checks, live OWASP ZAP scanning, UAT, and real-device checks are separate activities and are not established by these unit-test runs.

5. Issues observed during test execution

1. The initial client test attempt under Node.js 20.19.5 failed before tests could execute because Vitest workers could not start.
2. After switching the current PowerShell session to Node.js 22.22.2, all 6 client test files and 53 tests passed.
3. No failing tests remained in the final server or client runs.

6. Reproduction commands

```
cd server\nnpm test -- --runInBand\nnpm run test:coverage\n\ncd ..\\client\nnpm test -- --run
```

7. Conclusion

The recorded automated test run passed all 539 server and client tests. Server coverage was 95.16% for statements, 89.06% for branches, 94.90% for functions, and 95.99% for lines. This report records automated test evidence for the tested commit; it does not replace deployment security validation or user-acceptance testing.